

Secure Open-Source EDA

building trust into the tools that build silicon

Ishraq Tashdid

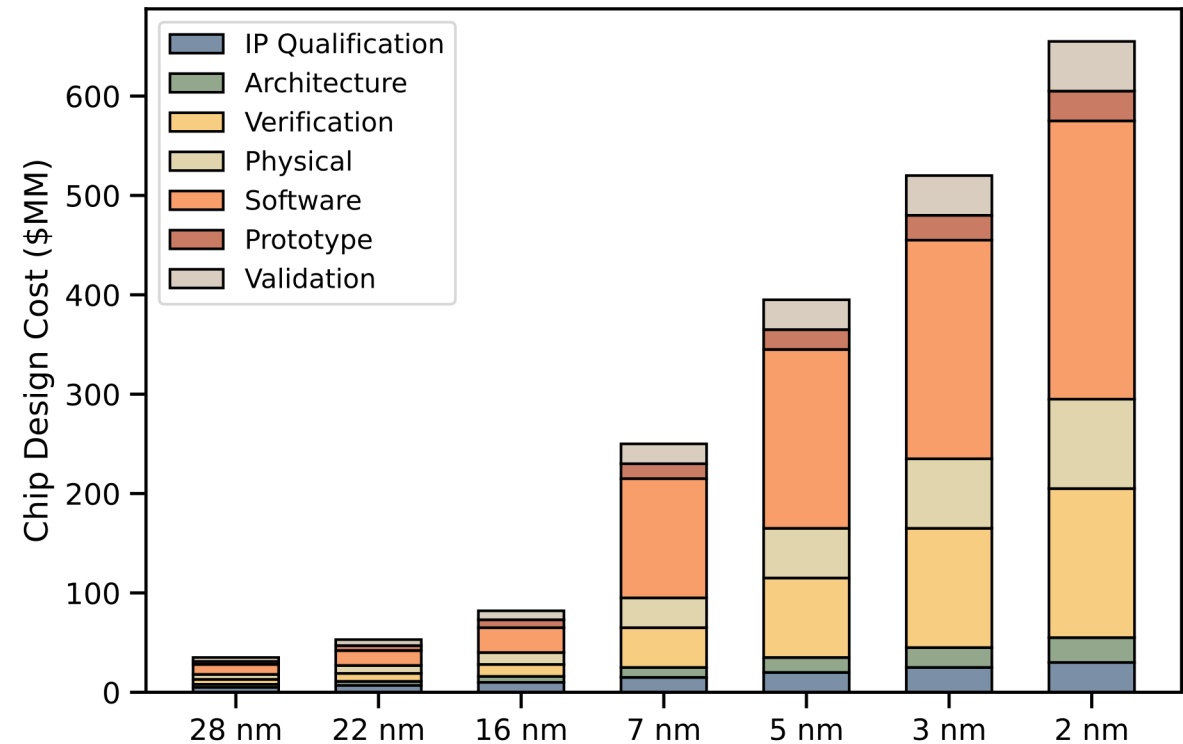
Security Researcher Intern Intel Corp
PhD Candidate at University of Central Florida, Orlando

DAC 2026
Birds-of-a-Feather Session

28 July 2026

Need for Open-Source EDA

- The semiconductor market is growing fast, on track for \$2.9T by 2029
- Every dollar of that growth runs through EDA tools at some point
- As nodes shrink, software licensing has become the largest cost in chip design
- Open-source EDA offers the same rigor, without the toll



Free · Community · Flexibility · Innovation · Accessibility

Adoption of Open-Source EDA

- Governments are putting real money behind it. The CHIPS R&D program and CHIPS Alliance alone represent \$13B
- DARPA is backing it too, through the POSH and IDEA programs
- Community efforts are keeping pace. The OpenROAD Initiative has grown from a research project into ORFS and OpenLane EDA
- And it's already producing results. RISC-V and OpenTitan show open-source silicon can compete, not just participate

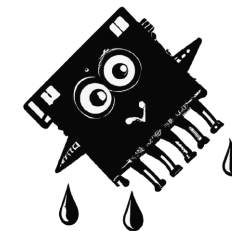


OpenROAD



Public Hardware Security Incidents

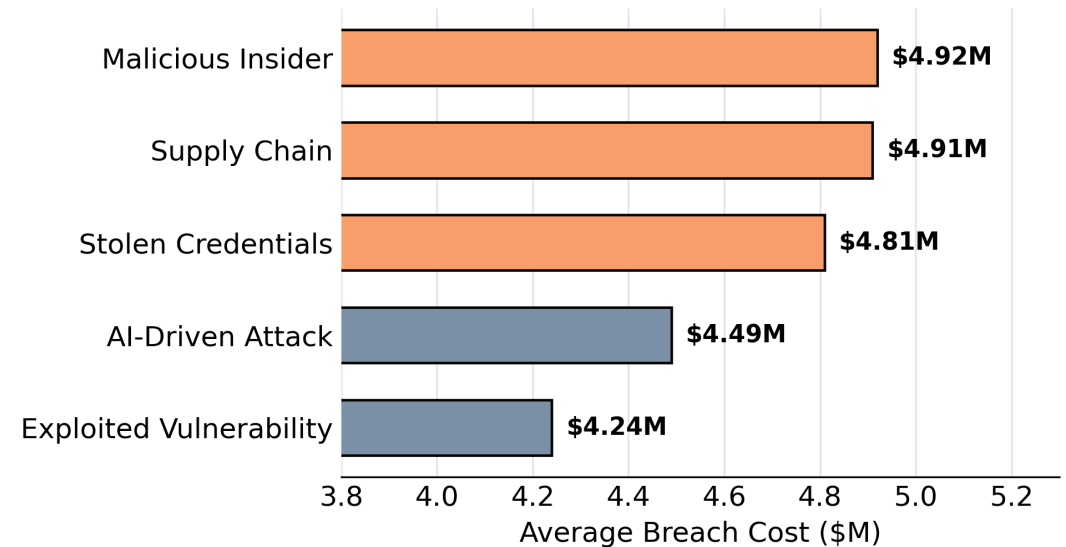
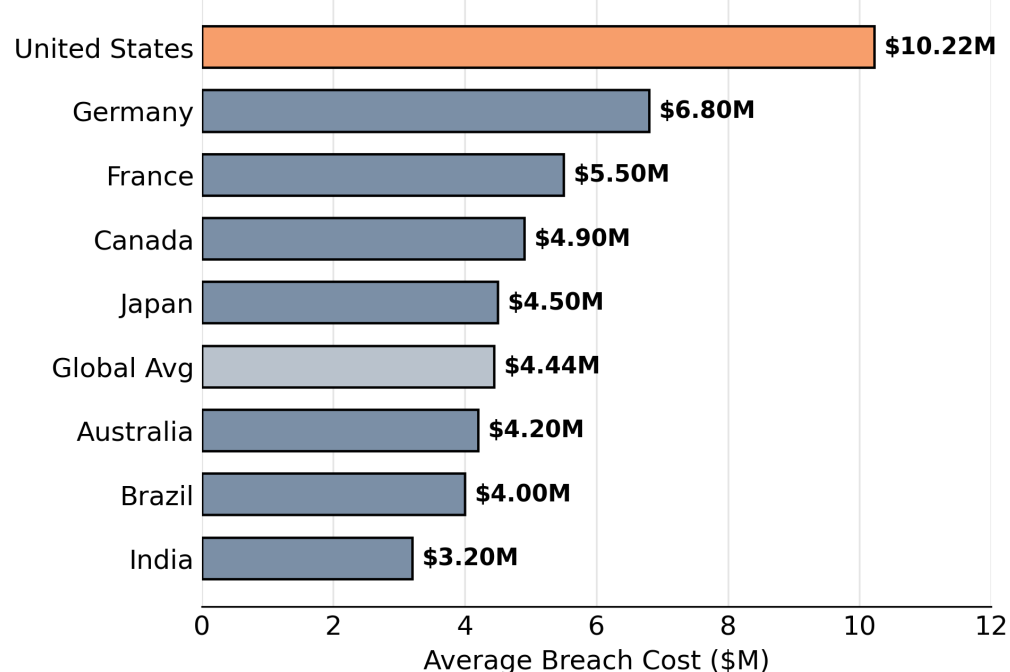
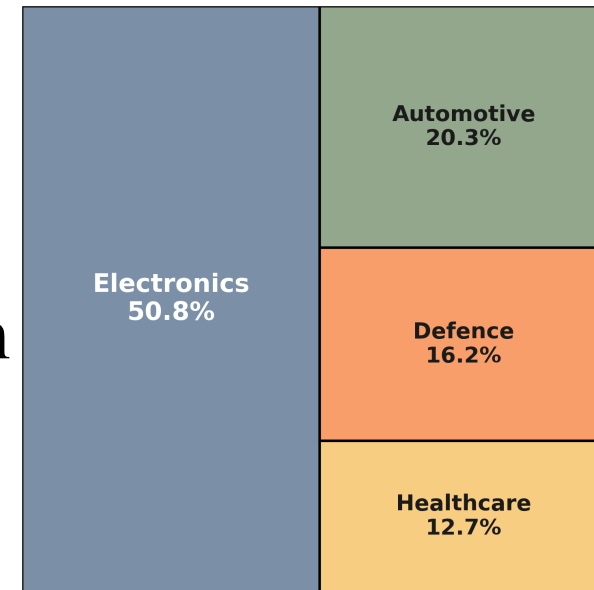
- Speculation & Side-Channel Leakage (Spectre)
 - Architectural checks may still “pass,” while transient microarchitectural behavior leaks sensitive data.
- Cross-Boundary Isolation Failure (Foreshadow)
 - A single flaw can impact multiple trust domains, from OS and VMs to enclave-style protected contexts.
- Stale Internal State Exposure (ZombieLoad)
 - Hidden microarchitectural buffers can unintentionally reveal leftover data across executions.
- Fault-Induced Security Breaks (Plundervolt)
 - Trusted system controls (like voltage/frequency tuning) can be turned into an attack path.
- Variant Recurrence in Related Mechanisms (Downfall)



Downfall

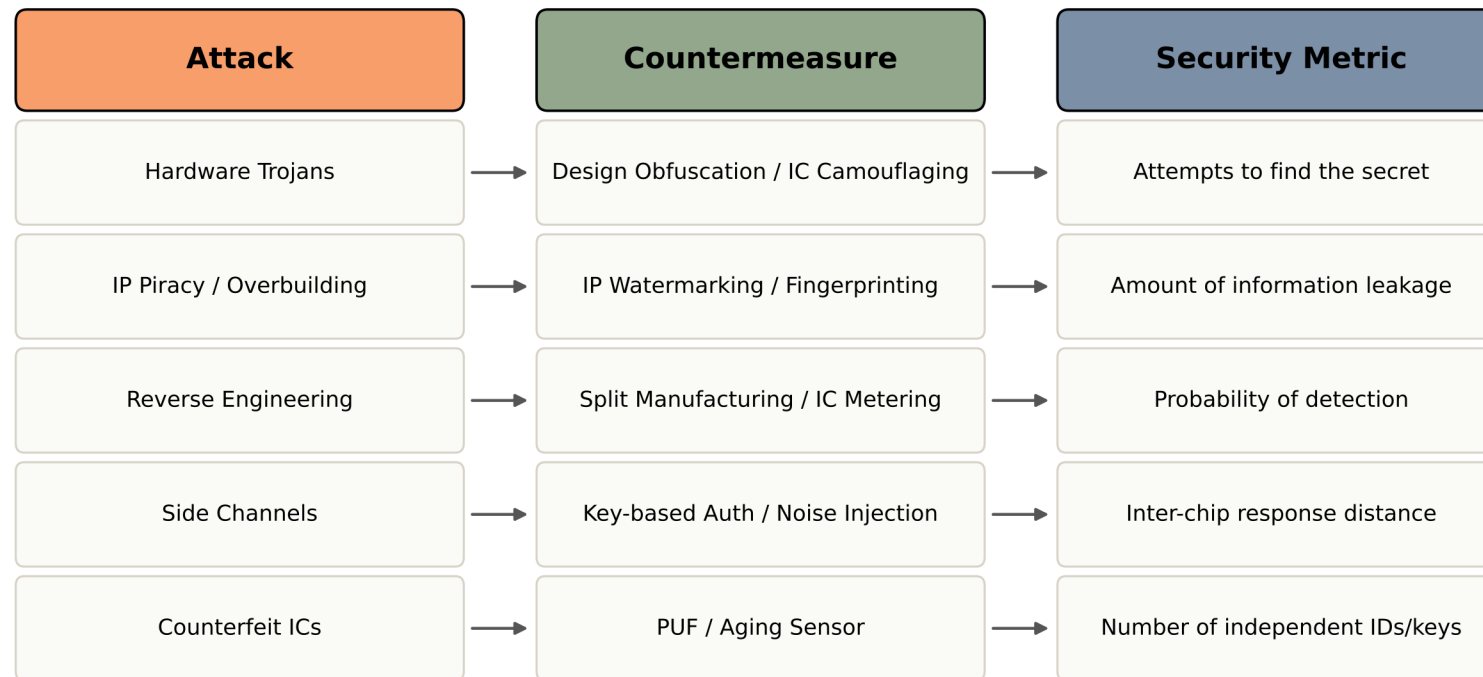
Impact of a Security Breach

- The global average breach now costs \$4.4M
- Breaches take 241 days on average to detect and contain
- The US has the highest cost of any country, an all-time high
- Third-party breaches have doubled
- Healthcare is the most targeted sector



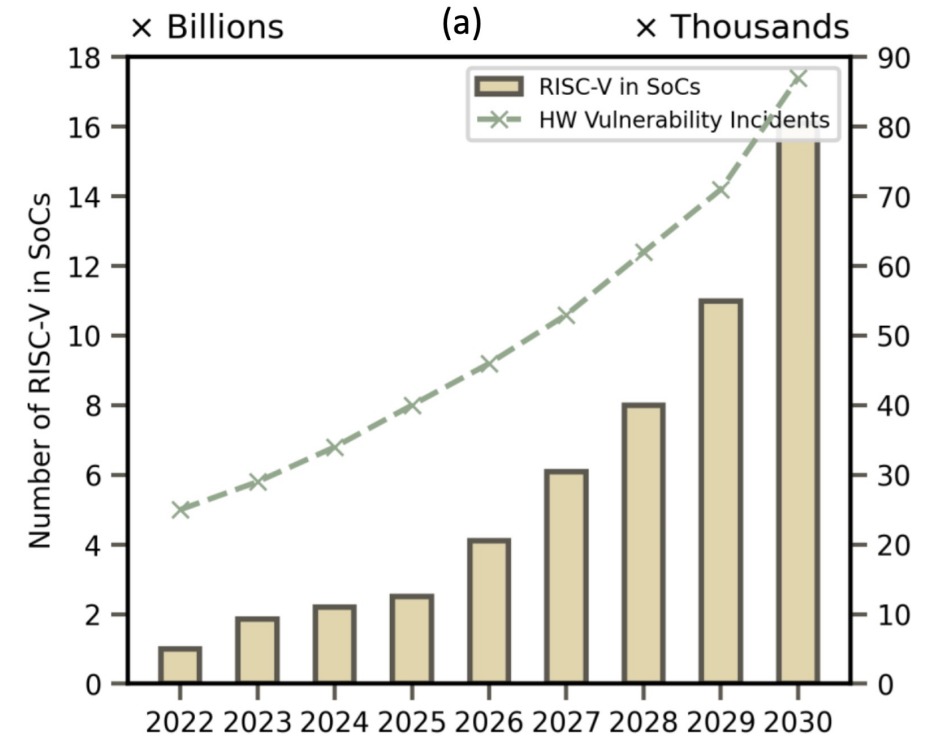
Quantifying Security Analysis

- Hardware security needs the same rigor as functional verification: assurance, not assumption
- Security metrics let us verify coverage, discover residual risk, and enable countermeasures across different attack types



Rising Vulnerabilities vs. Human Bandwidth

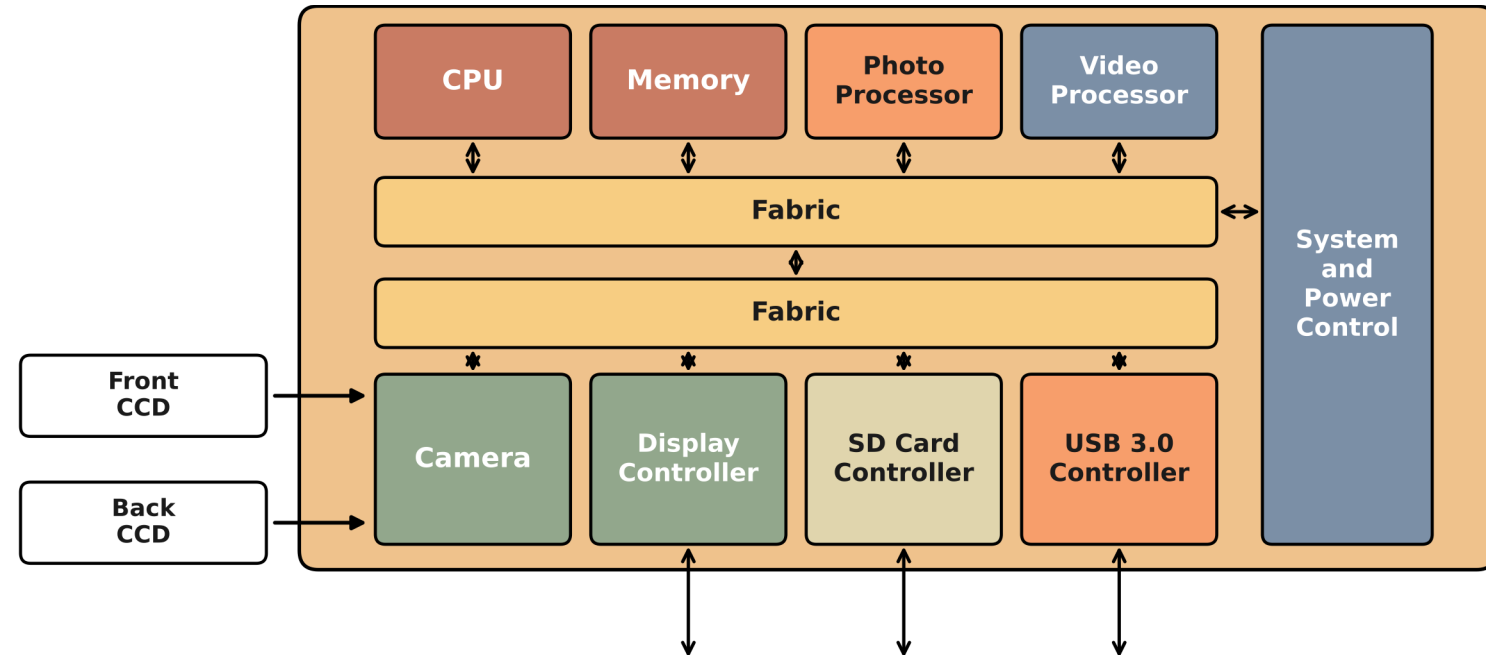
- Modern SoCs are growing in complexity, heterogeneity, and integration speed.
 - More IPs, more interfaces, more privilege modes, more corner cases.
- Security risk is expanding faster than teams can manually review and verify.
 - Threat analysis, property writing, and cross-module reasoning do not scale easily.
- In a fast-moving market, human expertise is essential but no longer sufficient alone.
 - We need workflows that amplify expert effort, not replace it.



Based on publicly reported SoC growth and hardware vulnerability incident trends (NIST/MITRE)

Countermeasure Feasibility

- Security vs. design constraints trade-off
- Coherence between countermeasures
- Successful threat mitigation
- Post-insertion functionality equivalence
- No PPA overhead violation
- Scalability to emerging threats



Conclusion

- Open-source EDA is no longer a research curiosity. It's backed by billions in funding, it spans the full RTL-to-GDSII flow, and it's already producing silicon that competes.
- That same openness is what makes it a target. Every stage from synthesis to physical design carries real, measurable risk, and the cost of getting it wrong is counted in the hundreds of billions.
- Security has to be designed in, not bolted on. The frameworks and metrics exist. What's needed now is for the community to treat security as a first-class requirement, the same way it already treats correctness and performance.

References

- WSTS, Global Semiconductor Market Surges Beyond \$1.5T (2026)
- TestFlow Inc., 2nm Chip Development Cost Analysis (\$725M)
- Panchal, Free and Open-Source VLSI Chip Design EDA Tools List, LinkedIn
- Integrated Circuit Design, Springer (2025)
- MDPI Electronics, Vol. 15, Issue 5, Art. 1048
- ACM Digital Library, DOI 10.1145/3505170.3511046
- VisionSpace, Remote Code Execution and Critical Vulnerabilities in NASA F Prime v3.4.3
- Solt et al., USENIX Security 2025
- Astute Group, Counterfeit Electronics Threaten Supply Chain Sustainability
- StationX, Cyber Security Breach Statistics
- DeepStrike, Supply Chain Attack Statistics 2025
- International Journal of Information Security, Springer, DOI 10.1007/s10207-026-01284-4
- IEEE Xplore, Document 6860363
- OpenROAD Initiative, Google Joins ORI, July 14, 2026
- Yahoo Finance, Cadence, Synopsys Slide on Kimi K3, July 16, 2026
- EuroCDP, Open EDA Tool Gap Analysis and Recommendations, July 10, 2026
- NSF Award #2449356

“The next chapter of open-source EDA won't be defined by whether it's adopted. It'll be defined by whether it's trusted.”